Customer Shopping Behaviour Analysis

A comprehensive data analysis and preprocessing report on 3,900 customer records spanning demographics, purchase patterns, preferences, and behavioral insights.

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Dataset Overview

3,900

Customer Records

Complete dataset with 18 original features

19

Final Features

After engineering and preprocessing

Data Structure

- 5 numeric columns (int64 and float64)
- 13 categorical columns (object type)
- Memory usage: ~548.6 KB

Key Features

- Demographics: Age, Gender, Location
- Purchase details: Amount, Category, Item
- Behavioral: Frequency, Reviews, Subscriptions

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Data Preprocessing Pipeline

1

Data Loading

Import CSV, inspect structure with Pandas

2

Data Cleaning

Handle missing values, remove duplicates

3

Feature Engineering

Create new features, standardize columns

4

Database Loading

Load to MySQL via SQLAlchemy

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Feature Engineering Highlights

Age Groups

Created 4 quantile-based segments: Young (18-30), Adult (31-43), Middle-age (44-56), Senior (57-70)

Purchase Frequency

Converted text labels to days: Weekly (7), Fortnightly (14), Monthly (30), Quarterly (90), Annually (365)

Column Standardization

Lowercase with underscores, removed redundant promo_code_used column

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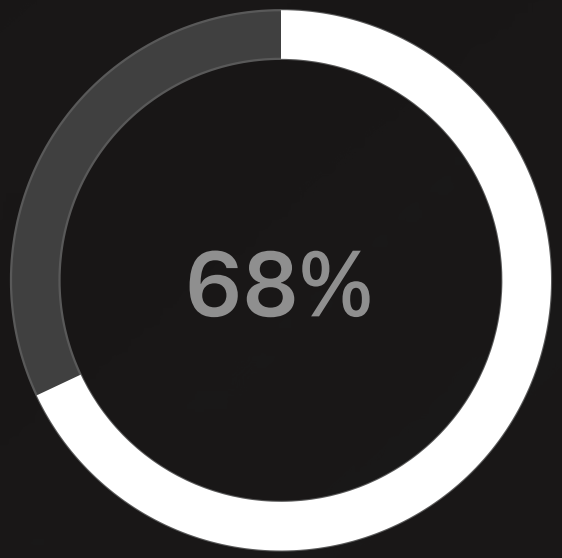
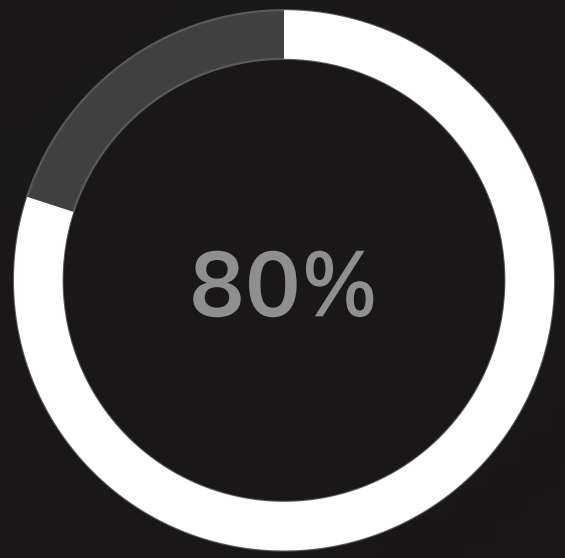
Key Business Insights



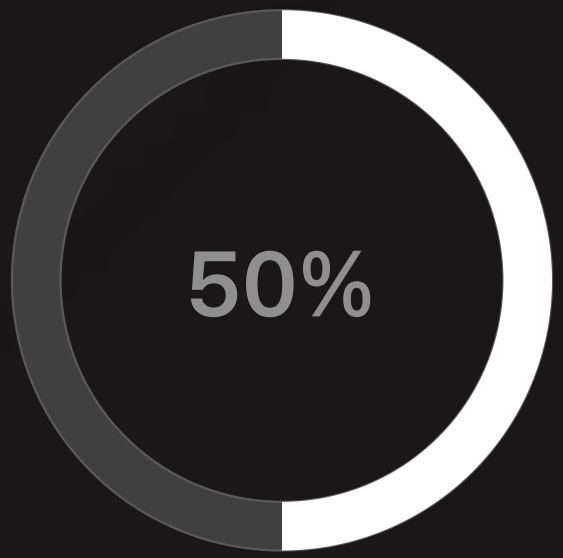
Male Revenue Share

$157,890 vs $75,191 female



Loyal Customers

3,116 customers with 11+ purchases



Hat Discount Rate

Top product with highest discount usage

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Product Performance Insights

Top Rated Products

1. Gloves: 3.86 average rating
2. Sandals: 3.84 average rating
3. Boots: 3.82 average rating
4. Hat: 3.80 average rating
5. Skirt: 3.78 average rating

Top Selling by Category

- Accessories: Jewelry (171 orders)
- Clothing: Pants & Blouse (171 each)
- Footwear: Sandals (160 orders)
- Outerwear: Jacket (163 orders)

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Customer Segmentation

New

83 customers (2.1%)

Returning

701 customers (18%)

Loyal

3,116 customers (80%)

Critical Opportunity: 72% of repeat buyers (2,518 customers) are not subscribed — representing a major upsell opportunity for subscription campaigns.

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Subscription & Discount Analysis

Subscription Revenue

Non-subscribers account for $170,436 (73% of total) vs $62,645 for subscribers. Average spend nearly identical ($59.87 vs $59.49).

High-Value Discount Users

839 customers used discounts AND spent above average ($59.76) — valuable targets for promotional campaigns.

Shipping Behavior

Express shipping customers spend $60.48 vs $58.46 for Standard — speed-priority customers willing to spend more.

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Tools & Technologies

Python & Pandas

Data manipulation, cleaning, and feature engineering

MySQL

Relational database for storage and querying

SQLAlchemy

ORM and database connection engine

Power BI

Visual dashboard for customer behavior insights

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